

第一章

複數平面

1-1

P.2

1. (1) 原式 = $2\sqrt{2}i + 2i$ (2) 原式 = $2\sqrt{2}i \cdot 2i = 4\sqrt{2}i^2 = -4\sqrt{2}$

(3) 原式 = $1 + (1 - 4\sqrt{3}i) = 2 - 4\sqrt{3}i = 2 + 4\sqrt{3}i$

類題 1.

(1) $\frac{\sqrt{6}}{\sqrt{-8}} = \frac{\sqrt{6}}{\sqrt{8}i} = -\frac{\sqrt{3}}{2}i$

(2) 原式 = $(1+i)^3 - \sqrt{7-\sqrt{12}} \cdot i = (-2+2i) - \sqrt{7-\sqrt{12}} \cdot i = -2 + (2-\sqrt{7-\sqrt{12}})i = -2 + \sqrt{3}i$

2. (1) 原式 = $i^{10} + i^{11} + i^{12} + (i^{13} + \dots + i^{80}) = i^2 + i^3 + i^4 = -i$
連續四個相加=0

(2) 原式 = $(-1)^{11} \cdot i^{49} \cdot i^{50} = (-1) \cdot i \cdot (-1) = i$

(3) 原式 = $\frac{7+3i}{2+5i} \cdot \frac{2-5i}{2-5i} + \frac{10-5i}{-3+4i} \cdot \frac{-3-4i}{-3-4i} = (1-i) + (-2-i) = -1-2i$

(4) 原式 = $((1+i)^2)^{10} = (2i)^{10} = -1024$

類題 2.

(1) 原式 = $(-2i)^2 + (2i)^3 = -4 - 8i$ (2) 原式 = $i + \frac{1}{i} = i + (-i) = 0$

P.3

3. $(b-4a) + (ab+4)i = (-2a-1) + (b+7)i \therefore \begin{cases} b-4a = -2a-1 \\ ab+4 = b+7 \end{cases}$ 得 $(a,b) = (2,3) \vee (-\frac{1}{2}, -2)$

類題 3.

左 $x^2\sqrt{2} + (2xy+y)\sqrt{3} = (x^2+y^2+x)i = 9\sqrt{2} + 10\sqrt{3} + 10i$ 得 $\begin{cases} x^2 = 9 \\ 2xy+y = 10 \\ x^2+y^2+x = 10 \end{cases} (x,y) = (-3,-2)$

P.4

4. 公式解 $x = \frac{-1 \pm \sqrt{-3}}{2} = \frac{-1 \pm \sqrt{3}i}{2}$

類題 4.

公式解 $x = \frac{-6 \pm \sqrt{-104}}{10} = \frac{-3 \pm \sqrt{26}i}{5}$

5. (1) $D = k^2 - 4(2k-3) > 0, k^2 - 8k + 12 > 0, (k-2)(k-6) > 0, k > 6 \vee k < 2$

(2) $D = k^2 - 4(2k-3) = 0, k^2 - 8k + 12 = 0, (k-2)(k-6) = 0, k = 2 \vee k = 6$

(3) $D = k^2 - 4(2k-3) \geq 0, k^2 - 8k + 12 \geq 0, (k-2)(k-6) \geq 0, k \geq 6 \vee k \leq 2$

(4) $D = k^2 - 4(2k-3) < 0, k^2 - 8k + 12 < 0, (k-2)(k-6) < 0, 2 < k < 6$

類題 5.

$D = 4a^2 - 4(2a+ab-3) > 0, a^2 - 2a^2 - ab + 3 > 0, -a^2 - ab + 3 > 0, 3 > a(a+b)$

$\therefore a, b \in \mathbb{N} \therefore a+b \geq 2 \therefore a(a+b) = 2 \therefore (a,b) = (1,1)$

6. 令 $t = x^2 - 3x$, 原式 $\Rightarrow t^2 + 2t - 8 = 0 \Rightarrow (t+4)(t-2) = 0$, $t = -4 \vee 2$

當 $t = -4$ 時, $x^2 - 3x + 4 = 0$, $x = \frac{3 \pm \sqrt{7}i}{2}$

當 $t = 2$ 時, $x^2 - 3x - 2 = 0$, $x = \frac{3 \pm \sqrt{17}}{2}$

類題 6.

$$[(x^2 + 8) + x][(x^2 + 8) - 10x] + 24x^2 = (x^2 + 8)^2 - 9(x^2 + 8)x + 14x^2$$

$$= [(x^2 + 8) - 7x][(x^2 + 8) - 2x] = 0 \quad \therefore x = \frac{7 \pm \sqrt{17}}{2} \vee 1 \pm \sqrt{7}i$$

P.5

7. 根與係數 $\begin{cases} \alpha + \beta = -4 \\ \alpha\beta = 6 \end{cases}$

(1) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = (-4)^2 - 2 \cdot 6 = 4$

(2) $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = (-4)^3 - 3 \cdot 6 \cdot (-4) = 8$

(3) $\alpha^4 + \beta^4 = (\alpha^2 + \beta^2)^2 - 2\alpha^2\beta^2 = 4^2 - 2 \cdot 6^2 = -56$

(4) 原式 $= \frac{(\alpha+3) + (\beta+3)}{(\alpha+3)(\beta+3)} = \frac{\alpha + \beta + 6}{\alpha\beta + 3(\alpha + \beta) + 9} = \frac{-4 + 6}{6 + 3(-4) + 9} = \frac{2}{3}$

類題 7.

根與係數 $\begin{cases} \alpha + \beta = 5 \\ \alpha\beta = 7 \end{cases}$

(1) 原式 $= \frac{\beta(\beta+1) + \alpha(\alpha+1)}{(\alpha+1)(\beta+1)} = \frac{\alpha^2 + \beta^2 + \alpha + \beta}{\alpha\beta + \alpha + \beta + 1} = \frac{(\alpha + \beta)^2 - 2\alpha\beta + (\alpha + \beta)}{\alpha\beta + \alpha + \beta + 1} = \frac{5^2 - 2 \cdot 7 + 5}{7 + 5 + 1} = \frac{16}{13}$

(2) $(\alpha^2 + \beta^2)(\alpha^3 + \beta^3) - \alpha^2\beta^2(\alpha + \beta) = [(\alpha + \beta)^2 - 2\alpha\beta][(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)] - \alpha^2\beta^2(\alpha + \beta)$
 $= (5^2 - 2 \cdot 7)(5^3 - 3 \cdot 7 \cdot 5) - 7^2 \cdot 5 = -25$

8. $\begin{cases} \alpha + \beta = 2 \\ \alpha\beta = \frac{3}{2} \end{cases}$

新方程式兩根和 $\frac{\beta+1}{\alpha} + \frac{\alpha+1}{\beta} = \frac{\alpha^2 + \beta^2 + \alpha + \beta}{\alpha\beta} = \frac{(\alpha + \beta)^2 - 2\alpha\beta + \alpha + \beta}{\alpha\beta} = 2$

兩根積 $\frac{\beta+1}{\alpha} \cdot \frac{\alpha+1}{\beta} = \frac{\alpha\beta + \alpha + \beta + 1}{\alpha\beta} = 3$, 故新方程式 $x^2 - 2x + 3 = 0$

類題 8.

$$\begin{cases} \alpha + \beta = -4 \\ \alpha\beta = -1 \end{cases} \quad \text{設} \quad \begin{cases} A = \frac{\beta}{\alpha+2} \\ B = \frac{\alpha}{\beta+2} \end{cases}$$

$$A+B = \frac{\beta}{\alpha+2} + \frac{\alpha}{\beta+2} = \frac{\alpha^2 + \beta^2 + 2(\alpha+\beta)}{4 + \alpha\beta + 2(\alpha+\beta)} = \frac{(\alpha+\beta)^2 + 2(\alpha+\beta) - 2\alpha\beta}{4 + 2(\alpha+\beta) + \alpha\beta} = \frac{(-4)^2 + 2(-4) - 2(-1)}{4 + 2(-4) + (-1)} = -2$$

$$A \times B = \frac{\beta}{\alpha+2} \times \frac{\alpha}{\beta+2} = \frac{\alpha\beta}{\alpha\beta + 4 + 2(\alpha+\beta)} = \frac{-1}{(-1) + 4 + 2(-4)} = \frac{1}{5}$$

$$x^2 + 2x + \frac{1}{5} = 0 \Rightarrow 5x^2 + 10x + 1 = 0$$

9. 同除以 $1+i$: $x^2 + \frac{1}{2}[(a+2) + (2-a)i]x + (-6+3i) = 0$, 令一實根為 k $\therefore$ 另一根為 $\frac{1}{k}(-6+3i)$

$$\therefore k + \frac{1}{k}(-6+3i) = -\frac{1}{2}[(a+2) + (2-a)i] \quad \therefore \begin{cases} k + \frac{-6}{k} = -\frac{a+2}{2} \\ \frac{3}{k} = -\frac{2-a}{2} \end{cases} \quad \text{得 } (a, k) = (-5, 1), \text{ 故 } (1, -6+3i)$$

類題 9.

設此實根 $k \in R$, k 代入 $\Rightarrow (k)^2 + (a+i)k + (3-3i) = 0$,

$$(k^2 + ak + 3) + (k-3)i = 0 \Rightarrow k = 3, a = -4, \text{ 原式} = x^2 + (-4+i)x + (3-3i) = 0$$

另一根為 $-(-4+i) - k = 1-i$

P.6

10. (1) $Z = \pm\sqrt{16-30i} = \pm\sqrt{16-2\sqrt{225}} = \pm(\sqrt{25}-\sqrt{-9}) = \pm(5-3i)$

《另解》設 $Z = a+bi$, $Z^2 = a^2 - b^2 + 2abi = 16-30i \quad \therefore \begin{cases} a^2 - b^2 = 16 \\ 2ab = -30 \end{cases}$

$$\therefore (a, b) = (5, -3) \vee (-5, 3) \quad \therefore \pm(5-3i)$$

(2) 公式解: $x = \frac{(1+i) \pm \sqrt{(1+i)^2 + 4(4-8i)}}{2} \Rightarrow x = \frac{(1+i) \pm \sqrt{16-30i}}{2} \Rightarrow x = \frac{(1+i) \pm (5-3i)}{2}$
 $\Rightarrow x = (3-i) \vee (-2+2i)$

類題 10.

$$x = \frac{(3+i) \pm \sqrt{(3+i)^2 - 4(8-i)}}{2} = \frac{(3+i) \pm \sqrt{-24+10i}}{2} = \frac{(3+i) \pm (1+5i)}{2} = (1-2i) \vee (2+3i)$$

類題 14.

設此純虛根 $ki, k \in R$, ki 代入 $\Rightarrow (ki)^3 + a(ki)^2 + 4(ki) + 1 = 0$,

$$-k^3i - ak^2 + 4ki + 1 = (1 - ak^2) + (4k - k^3)i = 0, \text{ 得 } \begin{cases} 1 - ak^2 = 0 \\ 4k - k^3 = k(4 - k^2) = 0 \end{cases}, k = 2 \vee -2 \vee 0 (\text{不合})$$

$$k = 2i \text{ 代入 } 1 - ak^2 = 0 \Rightarrow 1 - a \cdot (-4) = 0 \Rightarrow a = -\frac{1}{4}$$

設實根 α , 由根與係數 $2i + (-2i) + \alpha = a = -\frac{1}{4} \Rightarrow \alpha = -\frac{1}{4}$, 故三根為 $-\frac{1}{4}, \pm 2i$

15. $\begin{array}{ccccccccccc} | & | & | & | & | & | & | & | & | & | & | \\ -4 & -3 & -2 & -1 & 0 & 1 & 2 & 3 & 4 & & \end{array}$

$$\begin{array}{ccccccccccc} + & - & - & - & + & - & - & - & + & & \end{array}$$

答: $(-4, -3) \vee (-1, 0) \vee (0, 1) \vee (3, 4)$

類題 15.

x	-3	-2	-1	0	1	2
$f(x)$	-	+	+	-	+	+

由勘根定理知 $k = -3 \vee -1 \vee 0$

P.9

16.

x	$-\infty$	-2	-1	0	1	2	$+\infty$
$f(x)$	-	-	-	-	+	+	+

由勘根定理知有一根在 $0 \sim 1$ 之間

$$f(1.5) = -0.125, f(1.6) = 0.696$$

$$f(1.51) = -0.047049, f(1.52) = 0.031808$$

故可知此根在 $1.51 \sim 1.52$ 之間

類題 16.

$$x^3 = 2 \Rightarrow x = \sqrt[3]{2}, \text{ 由計算機可得 } x = \sqrt[3]{2} \approx 1.25992 \approx 1.26$$

習題 1-1

1. $i^{17 \times \frac{2+50}{2}} = i^{442} = -1$

2. $\frac{1}{3+i} \times \frac{3-i}{3-i} + \frac{1}{a+bi} = \frac{3}{5} \Rightarrow \frac{3}{10} - \frac{1}{10}i + \frac{1}{a+bi} = \frac{6}{10} \Rightarrow \frac{1}{a+bi} = \frac{3}{10} + \frac{1}{10}i$
 $\Rightarrow a+bi = \frac{10}{3+i} = \frac{10}{3+i} \times \frac{3-i}{3-i} = 3-i$

3. $\frac{10i}{1-2i+3i^2-4i^3} = \frac{10i}{1-2i-3+4i} = \frac{10i}{-2+2i} \times \frac{1+i}{1+i} = \frac{5}{2} - \frac{5}{2}i = \overline{a+bi} = a-bi$, 故 $(a, b) = (\frac{5}{2}, \frac{5}{2})$

4. $D = (4k)^2 - 4(6k+4) < 0 \Rightarrow 4k^2 - 6k - 4 < 0 \Rightarrow 2k^2 - 3k - 2 < 0 \Rightarrow (2k+1)(k-2) < 0$
 $\Rightarrow -\frac{1}{2} < k < 2$

5. $\begin{cases} \alpha + \beta = -4 \\ \alpha\beta = -6 \end{cases}$

(1) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 16 + 12 = 28$

(2) $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = (-4)^3 - 3 \times (-6) \times (-4) = -64 - 72 = -136$

(3) $\alpha^4 + \beta^4 = (\alpha^2 + \beta^2)^2 - 2(\alpha\beta)^2 = 28^2 - 2 \times (-6)^2 = 712$

$$6. \begin{cases} \alpha + \beta = -4 \\ \alpha\beta = 1 \end{cases} \therefore \alpha\beta < 0 \therefore \sqrt{\alpha} \cdot \sqrt{\beta} = -\sqrt{\alpha\beta}, \alpha + 2\sqrt{\alpha}\sqrt{\beta} + \beta = (\alpha + \beta) - 2\sqrt{\alpha\beta} = -4 - 2 = -6$$

$$7. \begin{cases} \alpha + \beta = 4 \\ \alpha\beta = 2 \end{cases}$$

$$\text{和} = \frac{\beta}{\alpha+2} + \frac{\alpha}{\beta+2} = \frac{\alpha^2 + \beta^2 + 2(\alpha + \beta)}{\alpha\beta + 2(\alpha + \beta) + 4} = \frac{(\alpha + \beta)^2 + 2(\alpha + \beta) - 2\alpha\beta}{4 + 2(\alpha + \beta) + \alpha\beta} = \frac{4 + 2 \times 4 - 2 \times 2}{4 + 2 \times 4 + 2} = \frac{10}{7}$$

$$\text{積} = \frac{\beta}{\alpha+2} \times \frac{\alpha}{\beta+2} = \frac{\alpha\beta}{\alpha\beta + 2(\alpha + \beta) + 4} = \frac{2}{4 + 2 \times 4 + 2} = \frac{1}{7} \quad \text{答: } 7x^2 - 10x + 1 = 0$$

$$8. \text{設三根 } 1+i, \alpha, \beta, \begin{cases} \text{三根和: } 1+i+\alpha+\beta=3 \\ \text{三根積: } (1+i) \cdot \alpha \cdot \beta = 2-2i \end{cases} \Rightarrow \begin{cases} \alpha + \beta = 2-i \\ \alpha\beta = \frac{2-2i}{1+i} \cdot \frac{1-i}{1-i} = -2i \end{cases}$$

以 α, β 為兩根的方程式為 $x^2 - (2-i)x - 2i = 0 \Rightarrow (x-2)(x+i) = 0$, 故另外兩根為 $2, -i$

$$9. \beta = \frac{3-i}{1-i} = 2+i, a = -[(1-i) + (2+i)] = -3$$

$$10. \Delta = (1-5i)^2 + 4(12+5i) = 24+10i,$$

$$\text{公式解 } x = \frac{-(1-5i) \pm \sqrt{24+10i}}{2} = \frac{-(1-5i) \pm \sqrt{24+2\sqrt{-25}}}{2} = \frac{-(1-5i) \pm (5+i)}{2} = 2+3i \vee -3+2i$$

$$11. f(3+2i) = f(\overline{3-2i}) = \overline{f(3-2i)} = \overline{-5i+6} = 5i+6$$

P.10

$$12. \because f(x) \in \mathbb{R} \therefore x = -1 \pm 2i, \therefore 4 \text{ 根和} = -3, 2 \text{ 根和} = -2 \therefore \text{另 } 2 \text{ 根和} = -1$$

$$\therefore f(x) = (x^2 + 2x + 5) \left(x^2 + x + \frac{6}{5} \right), \text{得 } (a, b) = (7, 5)$$

$$13. f(2) > 0, f(1) < 0, f(0) < 0, f(-1) > 0, f(-2) < 0$$

$$14. \because \deg f(x) = 3, \text{又 } f(x) \in \mathbb{R} \therefore \text{至少一實根} \therefore x = \pm i \vee k (k \in \mathbb{R})$$

$$15. \text{設矩形長} = x, \text{寬} = 12-x, \text{則面積} = x(12-x) = 40 \Rightarrow x^2 - 12x + 40 = 0, \Delta = (-12)^2 - 4 \cdot 40 < 0$$

x 無實數解, 故無法用長度 24 公尺的繩子圍成面積 40 平方公尺的矩形

$$16. (A) \because f(x) \in \mathbb{R} \quad (B) \because x = 1 \pm i \vee k (k \in \mathbb{R})$$

$$(C) f(x) - x = 0, \because \deg[f(x) - x] = 3 \therefore \text{必至少一實解}$$

$$(D) \because \deg f(x^3) = 9 \therefore \text{必至少一實根} \quad (E) \because f(x) \text{ 僅一實根} \therefore f(4) < 0$$

$$17. \text{長除法} \begin{array}{r} 1 \quad +a \quad +1 \quad \overline{) 1 \quad +2 \quad -2 \quad +k} \\ 1 \quad +a \quad +1 \quad \hline (2-a) \quad -3 \quad +k \\ (2-a) + (2a-a^2) + (2-a) \quad \hline 0 \end{array}$$

$$\text{可得 } -3 - (2a - a^2) = 0 \Rightarrow a^2 - 2a - 3 = 0 \Rightarrow (a-3)(a+1) = 0, a = 3 \text{ 或 } -1$$

因為 $g(x) = x^2 + ax + 1$ 有虛根, 故取 $a = -1$

$$\text{由長除法可知 } f(x) = (x^2 - x + 1)(x + 3), \text{故 } f(x) = 0 \Rightarrow x = \frac{1 \pm \sqrt{-3}}{2} \text{ 或 } -3$$

故選(1)(4)

$$18. (xy+3)+(3y-x)i=(-1)+(3-4x)i \quad \begin{cases} xy+3=-1 \\ 3y-x=3-4x \end{cases} \quad \begin{cases} xy=-4 \\ x+y=1 \end{cases}, \text{ 設 } x>y \therefore x>0>y$$

$$\frac{x}{y} + \frac{y}{x} + 2\sqrt{\frac{x}{y}} \cdot \sqrt{\frac{y}{x}} = \frac{x^2+y^2}{xy} + 2\sqrt{\frac{x}{-y}}i \times \sqrt{\frac{-y}{x}}i = \frac{(x+y)^2-2xy}{xy} - 2 = \frac{-17}{4}$$

$$19. \text{ 設 } a=x+yi, b=x-yi, \text{ 根與係數 } (x+yi)+(x-yi)=-1 \Rightarrow x=-\frac{1}{2}$$

$$\text{且 } a=\frac{1}{b} \Rightarrow -\frac{1}{2}+yi=\frac{1}{-\frac{1}{2}-yi}, \left(-\frac{1}{2}\right)^2+y^2=1, y^2=\frac{3}{4},$$

$$\text{所求 } a^2+b^2=(a+b)^2-2ab=(-1)^2-2(x^2+y^2)=-1$$

$$20. (x^2-1)(x^2+1)^2=0, x=\pm 1 \vee \pm i (\text{重根})$$

$$21. \because f(x) \in \mathbb{R} \therefore \overline{a+bi}=1+bi \therefore (a,b)=(1,-1), \therefore \alpha+\beta+\gamma=30=(1+i)(1-i) \cdot \gamma \therefore \gamma=15$$

$$22. f(x)^3+1=0 \Rightarrow f(x)^3=-1, f(x)=-1 \vee -\omega \vee -\omega^2$$

$$\textcircled{1} x^2+x+1<-1, D=1-4 \times 2 < 0$$

$$\textcircled{2} \textcircled{3} \text{ 設 } x \in \mathbb{R} \therefore x^2+x+1 \in \mathbb{R}, \text{ 但 } x^2+x+1=-\omega \vee -\omega^2 \notin \mathbb{R}, \text{ 故設錯了 } \therefore x \notin \mathbb{R}$$

P.11

$$23. \text{ 原式: } \frac{b_1(x-a_1)(x-a_3)+b_2(x-a_1)(x-a_3)+b_3(x-a_1)(x-a_3)}{(x-a_1)(x-a_2)(x-a_3)}=1$$

$$\Rightarrow \frac{b_1(x-a_1)(x-a_3)+b_2(x-a_1)(x-a_3)+b_3(x-a_1)(x-a_3)-(x-a_1)(x-a_2)(x-a_3)}{(x-a_1)(x-a_2)(x-a_3)}=0$$

$$\text{設 } f(x)=b_1(x-a_1)(x-a_3)+b_2(x-a_1)(x-a_3)+b_3(x-a_1)(x-a_3)-(x-a_1)(x-a_2)(x-a_3)$$

與原方程式的實根相同，由勘根定理

x	$-\infty$	a_1	a_2	a_3	∞
$f(x)$	+	+	-	+	-

可知 $f(x)=0$ 有三相異實根，故原方程式也有三相異實根

1-2

P.14

$$1. (1) |z_a - z_c| = |(2+3i) - (-1-i)| = |3+4i| = \sqrt{3^2+4^2} = 5$$

$$(2) \text{ 題意即 } z_b - z_a = z_c - z_d \Rightarrow -6-2i = (-1-i) - z_d \Rightarrow z_d = 5+i$$

類題 1.

$$z=1-2i, \bar{z}=1+2i, -z=-1+2i, -\bar{z}=-1-2i, \text{ 即在複數平面上}$$

$$A(1,-2), B(1,2), C(-1,2), D(-1,-2), ABCD \text{ 為長方形, 面積} = 2 \times 4 = 8$$

$$2. |z - (-2i)| = 3, \text{ 即 } z \text{ 到 } (0,-2) \text{ 的距離為 } 3, \text{ 也就是以圓心為 } (0,-2) \text{ 半徑 } 3 \text{ 的圓}$$

類題 2.

題意即 z 到 $(1,2)$ 的距離等於 z 到 $(-3,4)$ 的距離，也就是 $(1,2)$ 與 $(-3,4)$ 的中垂線

P.15

$$3. |z| = \frac{|2-7i||2-3i||3+4i|}{|2+7i||3+2i|} = \frac{\sqrt{53} \cdot \sqrt{13} \cdot 5}{\sqrt{53} \cdot \sqrt{13}} = 5$$

類題 3.

$$z_a = \frac{\sqrt{3}+i}{4}, z_b = 1-i, \text{ 所求} = |z_a|^2 |z_b|^5 |z_c|^5 = \frac{1}{4} \cdot (\sqrt{2})^5 \cdot \left(\frac{1}{\sqrt{2}}\right)^5 = \frac{1}{4}$$

P.16

4. (1) $5(\cos 270^\circ + i \sin 270^\circ)$ (2) $\sqrt{(1+\sqrt{3})^2 + (1-\sqrt{3})^2} \left(\frac{1+\sqrt{3}}{2\sqrt{2}} + \frac{1-\sqrt{3}}{2\sqrt{2}}i\right) = 2\sqrt{2}(\cos 345^\circ + i \sin 345^\circ)$

(3) $\cos 70^\circ + i \sin 70^\circ$ (4) $3(-\sin 57^\circ + i \cos 57^\circ) = 3(\cos 147^\circ + i \sin 147^\circ)$

(5) $z = 1 + \cos 140^\circ + i \sin 140^\circ = 1 + (2\cos^2 70^\circ - 1) + i(2\sin 70^\circ \cos 70^\circ)$
 $= 2\cos^2 70^\circ + i(2\sin 70^\circ \cos 70^\circ) = 2\cos 70^\circ(\cos 70^\circ + i \sin 70^\circ)$

類題 4.

(1) $\sqrt{12+4} \cdot \left(\frac{-2\sqrt{3}}{4} + \frac{-2}{4}i\right) = 4(\cos 210^\circ + i \sin 210^\circ)$

(2) $2i = 2(\cos 90^\circ + i \sin 90^\circ)$

(3) $\overline{-1+i} = -1-i = \sqrt{2}\left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i\right) = \sqrt{2}(\cos 225^\circ + i \sin 225^\circ)$

P.17

5. $\frac{Z}{Z-1} = 2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}) = 2\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = 1 + \sqrt{3}i$

$Z = (1 + \sqrt{3}i)(Z-1) \Rightarrow (1-1-\sqrt{3}i)Z = -1-\sqrt{3}i \Rightarrow Z = \frac{-1-\sqrt{3}i}{-\sqrt{3}i} = 1 + \frac{\sqrt{3}}{3}i$

類題 5.

$\because |Z_1| = |Z_2| = \left|\frac{Z_1}{Z_2}\right| = 1, \frac{Z_2}{Z_1} = 1(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}) = i, \frac{1+bi}{\sqrt{3}+ai} = i \Rightarrow (1+a) + (b-\sqrt{3})i = 0$

$(a, b) = \Rightarrow (-1, \sqrt{3})$

6. (1) $1 \cdot 1 \cdot \left[\cos\left(\frac{8}{7}\pi + \frac{13}{7}\pi\right) + i \sin\left(\frac{8}{7}\pi + \frac{13}{7}\pi\right)\right] = 1(\cos 3\pi + i \sin 3\pi), \text{ 所求} = \pi$

(2) $\frac{(\cos 317^\circ + i \sin 317^\circ)(\cos 137^\circ + i \sin 137^\circ)}{\sin 56^\circ + i \cos 56^\circ} = \frac{\cos 454^\circ + i \sin 454^\circ}{\cos 34^\circ + i \sin 34^\circ} = \cos 420^\circ + i \sin 420^\circ$

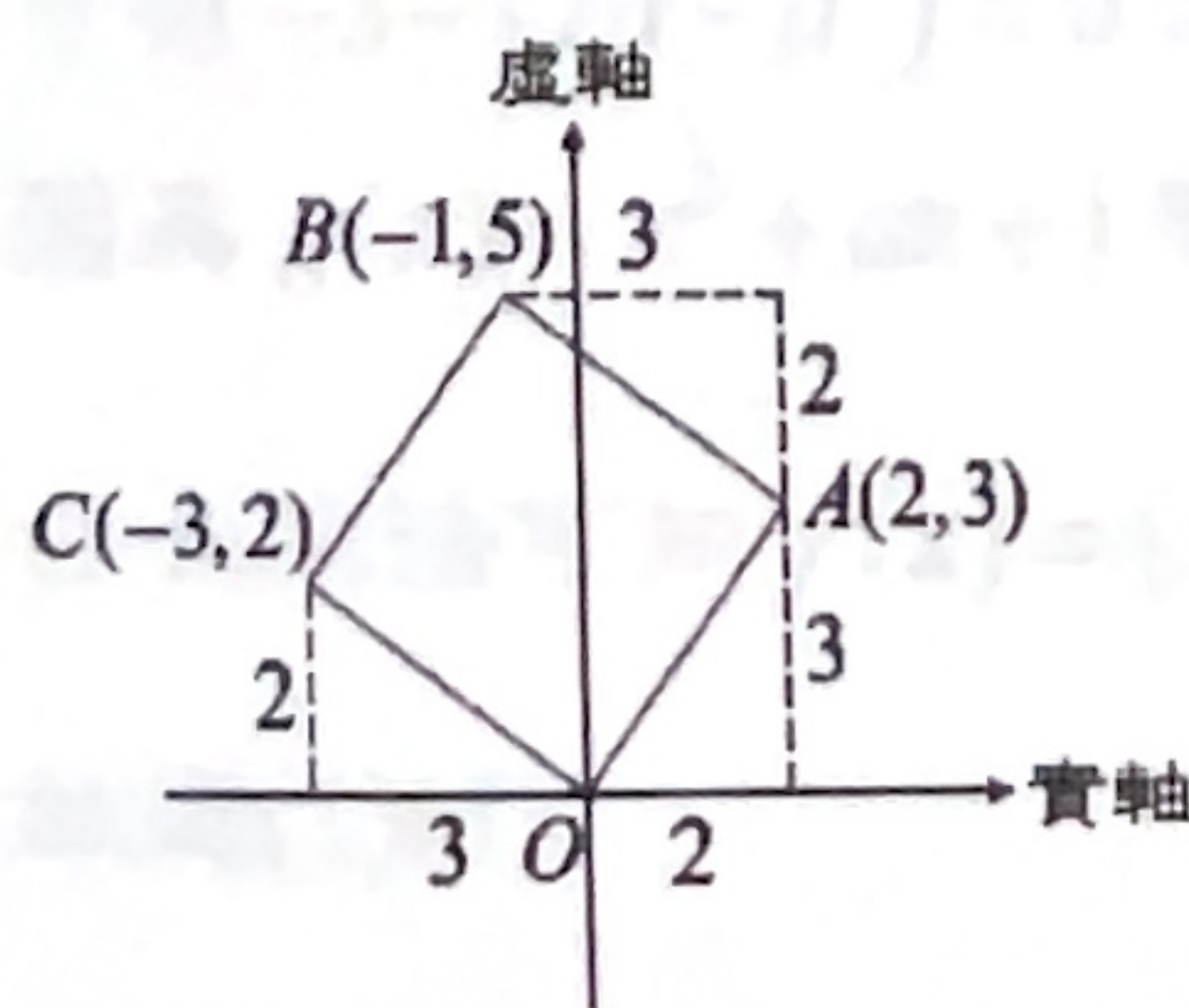
$= \frac{1+\sqrt{3}i}{2}, \text{ 所求} = \frac{7}{3}\pi - 2\pi = \frac{\pi}{3}$

類題 6.

上下同乘以 $\cos \frac{\pi}{8}$: $\frac{\cos \frac{\pi}{8} + i \sin \frac{\pi}{8}}{\cos \frac{\pi}{8} - i \sin \frac{\pi}{8}} = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$

P.18

7.



故 $B(-1+5i), C(-3+2i)$

類題 7.

$$A(x, y)$$

$$\begin{cases} (x-1)^2 + (y-2)^2 = 8 \\ (x-3)^2 + y^2 = 8 \end{cases} \Rightarrow \begin{cases} x^2 - 2x + 1 + y^2 - 4y + 4 = 8 \\ x^2 - 6x + 9 + y^2 = 8 \end{cases}$$

$$\Rightarrow \begin{cases} x^2 + y^2 - 2x - 4y = 3 \\ x^2 + y^2 - 6x = -1 \end{cases} \Rightarrow x - y = 1 \text{ 代回 } (x-3)^2 + y^2 = 8$$

$$(y-2)^2 + y^2 = 8 \Rightarrow 2y^2 - 4y - 4 = 0, \quad y^2 - 2y - 2 = 0 \Rightarrow y = \frac{2 \pm 2\sqrt{3}}{2} = 1 \pm \sqrt{3}$$

$$\text{答: } (2 + \sqrt{3}, (1 + \sqrt{3})i) \text{ 或 } (2 - \sqrt{3}, (1 - \sqrt{3})i)$$

$$8. (1) \left(\frac{\sqrt{3}+i}{\sqrt{2}}\right)^{30} = \sqrt{2} \left(\frac{\sqrt{3}}{2} + \frac{1}{2}\right)^{30} = \sqrt{2} (\cos 30^\circ + i \sin 30^\circ)^{30} = (\sqrt{2})^{30} (\cos 900^\circ + i \sin 900^\circ) \\ = 2^{15} (\cos 180^\circ + i \sin 180^\circ) = -2^{15}$$

$$(2) \frac{\left[2 \cdot \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)\right]^9}{\left[\sqrt{2} \cdot \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i\right)\right]^{10}} = \frac{(2)^9 (\cos 300^\circ + i \sin 300^\circ)^9}{(\sqrt{2})^{10} (\cos 45^\circ + i \sin 45^\circ)} = 2^4 (\cos 2250^\circ + i \sin 2250^\circ) \\ = 2^4 (\cos 90^\circ + i \sin 90^\circ) = 16i$$

類題 8.

$$\text{原式} = \frac{(2(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}))^7}{(\sqrt{2}(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}))^{10}} = \frac{2^7}{\sqrt{2}^{10}} (\cos \frac{10}{3}\pi + i \sin \frac{10}{3}\pi) = 4(-\frac{1}{2} - \frac{\sqrt{3}}{2}i) = -2 - 2\sqrt{3}i$$

$$9. \begin{cases} Z = \cos \theta + i \sin \theta \\ \frac{1}{Z} = \cos(-\theta) + i \sin(-\theta) \end{cases}, \quad Z + \frac{1}{Z} = 2 \cos \theta = \sqrt{3} \Rightarrow \cos \theta = \frac{\sqrt{3}}{2} \therefore \theta = \frac{\pi}{6} \text{ 或 } \frac{11\pi}{6}$$

$$\theta = \frac{\pi}{6} \text{ 時, } Z^9 = \cos(9 \cdot \frac{\pi}{6}) + i \sin(9 \cdot \frac{\pi}{6}) = -i, \quad \frac{1}{Z^{12}} = \cos(12 \cdot \frac{\pi}{6}) + i \sin(12 \cdot \frac{\pi}{6}) = 1, \quad Z^9 + \frac{1}{Z^{12}} = 1 - i$$

$$\theta = \frac{11\pi}{6} \text{ 時, } Z^9 = \cos(\frac{11\pi}{6} \cdot 9) + i \sin(\frac{11\pi}{6} \cdot 9) = i, \quad \frac{1}{Z^{12}} = \cos(-\frac{11\pi}{6} \cdot 12) + i \sin(-\frac{11\pi}{6} \cdot 12) = 1$$

$$Z^9 + \frac{1}{Z^{12}} = 1 + i \quad \text{答: } \underline{1 \pm i}$$

類題 9.

$$\begin{cases} Z = \cos \theta + i \sin \theta \\ \frac{1}{Z} = \cos(-\theta) + i \sin(-\theta) \end{cases}, \quad Z + \frac{1}{Z} = 2 \cos \theta = 1 \Rightarrow \cos \theta = \frac{1}{2} \therefore \theta = \frac{\pi}{3} \text{ 或 } \frac{5\pi}{3}$$

$$\textcircled{1} \theta = \frac{\pi}{3} \text{ 時, } Z^{101} = \cos(\frac{\pi}{3} \cdot 101) + i \sin(\frac{\pi}{3} \cdot 101) = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$\frac{1}{Z^{101}} = \cos(-\frac{\pi}{3} \cdot 101) + i \sin(-\frac{\pi}{3} \cdot 101) = \frac{1}{2} - \frac{\sqrt{3}}{2}i \Rightarrow Z^{101} + \frac{1}{Z^{101}} = 1$$

$$\textcircled{2} \theta = \frac{5\pi}{3} \text{ 時, } Z^{101} = \cos(\frac{5\pi}{3} \cdot 101) + i \sin(\frac{5\pi}{3} \cdot 101) = \frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad \frac{1}{Z^{101}} = \frac{1}{2} - \frac{\sqrt{3}}{2}i, \quad Z^{101} + \frac{1}{Z^{101}} = 1$$

P. 19

$$\begin{aligned}
 10. (1) \quad Z &= \left(\frac{\sqrt{3}+i}{2}\right)^7 + 1 = \left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)^7 + 1 = \cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6} + 1 \\
 &= \left(2\cos^2\frac{7\pi}{12} - 1\right) + i\left(\sin\frac{7\pi}{12}\cos\frac{7\pi}{12}\right) + 1 = 2\cos\frac{7\pi}{12}\left(\cos\frac{7\pi}{12} + i\sin\frac{7\pi}{12}\right) \\
 \therefore \cos\frac{7\pi}{12} &= -\cos\frac{5\pi}{12}, \quad Z = 2\cos\frac{5\pi}{12}\left(-\cos\frac{7\pi}{12} + i\sin\frac{7\pi}{12}\right) = 2\cos\frac{5\pi}{12}\left(\cos\frac{19\pi}{12} + i\sin\frac{19\pi}{12}\right) \\
 |Z| &= 2\cos\frac{5}{12}\pi = 2 \cdot \cos 75^\circ = \frac{\sqrt{6}-\sqrt{2}}{2}, \quad \text{Arg} Z = \frac{19}{12}\pi
 \end{aligned}$$

$$(2) \quad Z = 2 \cos \frac{5\pi}{12} \left(\cos \frac{19\pi}{12} + i \sin \frac{19\pi}{12} \right), \quad Z^n = \left(2 \cos \frac{5\pi}{12} \right)^n \left(\cos \frac{19n\pi}{12} + i \sin \frac{19n\pi}{12} \right)$$

$$\therefore Z^n \in \mathbb{R} \Rightarrow \frac{19}{12}n\pi \text{ 是 } \pi \text{ 的整數倍, } n=12, 24, 36, 48, 60, 72, 84, 96 \text{ 共 } \underline{8} \text{ 個}$$

類題 10.

$$\begin{aligned}\text{原式} &= \frac{1+2\cos^2 10^\circ-1+i(2\sin 10^\circ\cos 10^\circ)}{1+2\cos^2 10^\circ-1-i(2\sin 10^\circ\cos 10^\circ)} = \frac{2\cos 10^\circ(\cos 10^\circ+i\sin 10^\circ)}{2\cos 10^\circ(\cos 10^\circ-i\sin 10^\circ)} \\ &= \frac{\cos 10^\circ+i\sin 10^\circ}{\cos(-10^\circ)+i\sin(-10^\circ)} = \cos 20^\circ+i\sin 20^\circ\end{aligned}$$

$$Z^n = \cos(20^\circ \cdot n) + i \sin(20^\circ \cdot n), \because Z^n \in \mathbb{R}, 20^\circ \cdot n \text{ 是 } 180^\circ \text{ 的整數倍, 最小 } n = 9$$

P.20

11. $x^4 = -1 = \cos(\pi + 2k\pi) + i\sin(\pi + 2k\pi)$, $x = \cos \frac{\pi + 2k\pi}{4} + i\sin \frac{\pi + 2k\pi}{4}, k = 0, 1, 2, 3$

類題 11.

$$x^6 = -1 = \cos(\pi + 2k\pi) + i \sin(\pi + 2k\pi) \cdot x = \cos \frac{(2k+1)\pi}{6} + i \sin \frac{(2k+1)\pi}{6}, k = 0, 1, 2, 3, 4, 5$$

12. 設 $z^3 = -2 + 2i = 2\sqrt{2}(-\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}) = 2\sqrt{2}[\cos(\frac{3}{4}\pi + 2k\pi) + i\sin(\frac{3}{4}\pi + 2k\pi)]$

則 $z = \sqrt{2}[\cos(\frac{\pi}{4} + \frac{2k\pi}{3}) + i\sin(\frac{\pi}{4} + \frac{2k\pi}{3})], k = 0, 1, 2$

類題 12.

$$[Z - (-1 - 2i)]^4 = \sqrt{64 + 192} \left(\frac{-8}{16} + \frac{8\sqrt{3}}{16}i \right), \quad [Z - (-1 + 2i)]^4 = 16 \left(\underset{=\cos\theta}{\frac{-1}{2}} + \underset{=\sin\theta}{\frac{\sqrt{3}}{2}}i \right) = 16 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$$

$$Z - (1 - 2i) = 2 \left[\cos \left(\frac{\pi}{6} + \frac{k\pi}{2} \right) + i \sin \left(\frac{\pi}{6} + \frac{k\pi}{2} \right) \right], \quad k = \underline{0, 1, 2, 3}$$

P.21

13. $[Z - (-1 + 2i)]^4 = 2 + 2\sqrt{3}i$, $[Z - (-1 + 2i)]^4 = 4(\frac{2}{4} + \frac{2\sqrt{3}}{4}i)$, $[Z - (-1 + 2i)]^4 = 4(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3})$
 $Z - (-1 + 2i) = \sqrt[4]{4} \left[\cos(\frac{\pi}{12} + \frac{k\pi}{2}) + i\sin(\frac{\pi}{12} + \frac{k\pi}{2}) \right]$, $k = 0 \sim 3$

(1) 圓心坐標 $(-1, 2)$ — (2) 所求面積 $= (\sqrt[4]{4})^2 \cdot \frac{1}{2} \cdot 4 = \underline{4}$

類題 13.

$$Z^6 - 1 = (Z + 1)(Z^5 + Z^4 + Z^3 + Z^2 + Z + 1) = 0, \quad Z^6 = 1 \Rightarrow Z^6 = r^6(\cos 6\theta + i \sin 6\theta)$$

$$\begin{cases} r^6 = 1 \\ 6\theta = 2k\pi, k \text{ 為整數} \end{cases} \Rightarrow \begin{cases} r = 1 \\ \theta = \frac{k}{3}\pi \end{cases}$$

$$Z = \cos \frac{k\pi}{3} + i \sin \frac{k\pi}{3}, \quad k = 0 \sim 5$$

$Z + 1 = 0$ 之解不是原方程式解

$$\text{所求面積} = 4 \cdot \frac{\sqrt{3}}{2} + 2 \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{\sqrt{3}}{2} = \sqrt{3} + \frac{\sqrt{3}}{4} = \frac{5}{4}\sqrt{3}$$

習題 1-2

$$1. \quad |z| = \frac{|1+i|^3 |a-i|^2}{\sqrt{2} |a-3i|^2} = \frac{(\sqrt{2})^3 (a^2+1)}{\sqrt{2} \cdot (a^2+9)} = \frac{2}{3} \Rightarrow \frac{a^2+1}{a^2+9} = \frac{1}{3}, \quad 3a^2+3 = a^2+9 \Rightarrow a^2 = 3 \Rightarrow a = \pm\sqrt{3}$$

$$2. \quad (1) \quad \sqrt{9+27} \left(\frac{-3}{6} + \frac{3\sqrt{3}}{6}i \right) = 6(\cos 120^\circ + i \sin 120^\circ)$$

$$(2) \quad 3 \left(\frac{0}{\cos \theta} - \frac{\pi}{\sin \theta} \right) = 3(\cos 270^\circ + i \sin 270^\circ)$$

$$3. \quad Z = \frac{-2\sqrt{3}+2i}{\sqrt{3}+i} = \frac{(-2\sqrt{3}+2i)(\sqrt{3}-i)}{3-(-1)} = \frac{-6+2+4\sqrt{3}i}{4} = -1+\sqrt{3}i$$

$$Z = 2\left(\frac{-1}{2} + \frac{\sqrt{3}}{2}i\right) = 2\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right), \quad \text{Arg}(Z) = \frac{2}{3}\pi$$

$$4. \quad \overline{OA} = 3\sqrt{5}(\cos \alpha + i \sin \alpha), \quad \overline{OC} = \frac{1}{3}\overline{OA}$$

$$\overline{OC} = \overline{OA} \cdot \frac{1}{3}(\cos 90^\circ + i \sin 90^\circ) = \sqrt{5}[\cos(90^\circ + \alpha) + i \sin(90^\circ + \alpha)] = \sqrt{5} \cdot [-\sin \alpha + i \cos \alpha]$$

$$= \sqrt{5} \cdot \left[-\frac{1}{\sqrt{5}} + i \frac{2}{\sqrt{5}} \right] = -1 + 2i \quad \therefore C = (-1, 2), \quad \overline{OB} = \frac{\sqrt{10}}{3}\overline{OA}, \quad \text{轉}\beta$$

$$\therefore \overline{OB} = 5\sqrt{2}[\cos(\alpha + \beta) + i \sin(\alpha + \beta)]$$

$$\text{其中 } \cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta = \frac{2}{\sqrt{5}} \cdot \frac{3}{\sqrt{10}} - \frac{1}{\sqrt{5}} \cdot \frac{1}{\sqrt{10}} = \frac{1}{\sqrt{2}}$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta = \frac{1}{\sqrt{5}} \cdot \frac{3}{\sqrt{10}} + \frac{2}{\sqrt{5}} \cdot \frac{1}{\sqrt{10}} = \frac{1}{\sqrt{2}}$$

$$\therefore \overline{OB} = 5 + 5i, \quad B = (5, 5)$$

$$5. \quad \begin{cases} a+c=0 \\ b+d=0 \end{cases}, \quad \begin{cases} m_{AC} \cdot m_{BD} = -1 \Rightarrow \frac{b-d}{a-c} \cdot \frac{6}{2} = -1 \dots \textcircled{1} \\ a^2 + b^2 = (-3)^2 + (-1)^2 = 10 \dots \textcircled{2} \end{cases}, \quad \textcircled{1} \text{式可整理為 } \frac{b+b}{a+a} = -3 \Rightarrow b = -3a \text{ 代入 } \textcircled{2}$$

$$\Rightarrow 10a^2 = 10 \Rightarrow a = \pm 1 \text{ (正的與圖不合)}, \quad (a, b, c, d) = (-1, 3, 1, -3), \quad \text{答: } \underline{B(-1+3i), D(1-3i)}$$

$$6. \overline{OB} = \overline{OA} [\cos(-150^\circ) + i \sin(-150^\circ)] = 6(\cos 30^\circ + i \sin 30^\circ) [\cos(-150^\circ) + i \sin(-150^\circ)]$$

$$= 6(\cos -120^\circ + i \sin -120^\circ) = 6(\cos 240^\circ + i \sin 240^\circ), \tan 240^\circ = \sqrt{3}$$

$$7. \cos 343^\circ + i \sin 163^\circ = \cos 17^\circ + i \sin 17^\circ, \sin 83^\circ + i \cos 83^\circ = \cos 7^\circ + i \sin 7^\circ$$

$$\text{所求} = \frac{(\cos 102^\circ + i \sin 102^\circ)(\cos 35^\circ + i \sin 35^\circ)}{\cos 13^\circ - i \sin 13^\circ} = \frac{\cos 137^\circ + i \sin 137^\circ}{\cos 347^\circ + i \sin 347^\circ} = \cos(-210^\circ) + i \sin(-210^\circ)$$

$$-\frac{\sqrt{3}}{2} + \frac{1}{2}i$$

$$8. Z = \frac{\sqrt{2}(\cos 315^\circ + i \sin 315^\circ)}{2(\cos 30^\circ + i \sin 30^\circ)} = \frac{1}{\sqrt{2}}(\cos 285^\circ + i \sin 285^\circ)$$

$$Z^n = \left(\frac{1}{\sqrt{2}}\right)^n (\cos 285^\circ n + i \sin 285^\circ n) \Rightarrow 285^\circ \cdot n \text{ 爲 } k\pi, n_{\min} \text{ 爲 } 12$$

$$1 \leq n \leq 100, \text{ 所有 } n \text{ 爲 } 12 \cdot 24 \cdot 36 \cdot 48 \cdot 60 \cdot 72 \cdot 84 \cdot 96 \Rightarrow \frac{(12+96) \cdot 8}{2} = 432$$

P.22

$$9. Z^6 = 1, Z = 1(\cos \frac{2k\pi}{6} + i \sin \frac{2k\pi}{6})$$

$$(A) \bigcirc : |Z| = 1 \quad (B) \times : Z^2 = \cos \frac{2}{3}\pi + i \sin \frac{2}{3}\pi = \frac{-1 + \sqrt{3}i}{2} \quad (C) \bigcirc : (Z^3)^2 = 1 \Rightarrow Z^3 = \pm 1$$

$$(D) \bigcirc : |Z^4| = |Z|^4 = 1 \quad (E) \bigcirc : Z^6 - 1 = (Z-1)(Z^5 + Z^4 + Z^3 + Z^2 + Z + 1) = 0 \dots \text{選}$$

$$(A)(C)(D)(E)$$

$$10. x^6 = i = \cos 90^\circ + i \sin 90^\circ$$

$$\therefore x_k = \cos(15^\circ + 60^\circ k) + i \sin(15^\circ + 60^\circ k), k = 0, 1, 2, 3, 4, 5$$

$$\text{而 } a+bi \text{ 之 } a < 0, b > 0, \text{ 第二象限} \Rightarrow x_2$$

$$\text{則 } \begin{cases} a = \cos 135^\circ = -\frac{\sqrt{2}}{2} \\ b = \sin 135^\circ = \frac{\sqrt{2}}{2} \end{cases} \Rightarrow ab = -\frac{1}{2}$$

$$11. \sqrt{1024 + 3072i} \left(\frac{32}{64} - \frac{32}{64}\sqrt{3}i \right) = 64 \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i \right)$$

$$Z^6 = 64 [\cos(300^\circ + 2k\pi), \sin(300^\circ + 2k\pi)]$$

$$Z = 2 \left[\cos(50^\circ + \frac{k\pi}{3}), \sin(50^\circ + \frac{k\pi}{3}) \right], k = 0 \sim 5$$

$$\text{所求面積} = 6 \cdot \left(\frac{1}{2} \cdot 2 \cdot 2 \cdot \sin 60^\circ \right) = 12 \cdot \frac{\sqrt{3}}{2} = 6\sqrt{3}$$

$$12. x^3 + 3x^2 + 3x + 28 \Rightarrow (x+1)^3 + 27 = 0 \therefore (x+1)^3 = -27 = 3^2(\cos 180^\circ + i \sin 180^\circ)$$

$$\Rightarrow x = -1 + 3 \left\{ [\cos(60^\circ + 120^\circ)] + [i \sin(60^\circ + 120^\circ)] \right\}, k = 0, 1, 2$$

$$k = 0: \frac{1-3\sqrt{3}i}{2}, k = 1: -4, k = 2: \frac{1+3\sqrt{3}i}{2}$$

$$13. z^5 = \cos 2\pi + i \sin 2\pi = 1, z^5 - 1 = (z-1)(z^4 + z^3 + z^2 + z + 1) = 0,$$

即 $z^4 + z^3 + z^2 + z + 1 = 0$ 連續五個相加為 0，所求 $= z^{65} = \cos 26\pi + i \sin 26\pi = 1$

$$14. \text{令 } Z = a \left(\cos \theta + i \sin \theta \right), a > 1 \Rightarrow \frac{1}{Z} = \frac{1}{a} \left(\cos -\theta + i \sin -\theta \right), \textcircled{1} \frac{1}{a} < 1 \quad \textcircled{2} \text{第四象限} \cdots \cdots \text{選(D)}$$

$$15. W = Z^3 = r^3 (\cos 3\theta + i \sin 3\theta) \text{ 而 } \frac{9\pi}{4} \leq \theta \leq \frac{15\pi}{4} \Rightarrow \text{所求}(e)$$

$$16. \text{令 } x = i + 2i^2 + 3i^3 + \cdots + 100i^{100}, x_i = i^2 + 2i^3 + \cdots + 99i^{100} + 100i^{101}$$

$$\therefore x - x_i = i + i^2 + i^3 + \cdots + i^{100} + 100i^{101} = 100i \quad \therefore x = \frac{100i}{1-i} = \frac{100(1+i)}{2} = 50 + 50i$$

$$\therefore |50 + 50i| = 50\sqrt{2}$$

$$17. (1) \text{題意即 } P(4, 5), Q(-4, 15), A(a, 0), \text{欲求 } \overline{AP} + \overline{AQ} \text{ 最小值}$$

$$\text{先將 } P(4, 5) \text{ 以軸成對稱得 } P'(4, -5), \text{所求} = \overline{P'Q} = \sqrt{8^2 + 20^2} = \sqrt{464}$$

$$(2) \text{此時, } A(a, 0) \text{ 爲 } \overleftrightarrow{P'Q}: 5x + 2y = 10 \text{ 與 } x \text{ 軸交點 } (2, 0), a = 2$$

$$18. \frac{(\cos A - i \sin A)(\cos B + i \sin B)}{\cos C - i \sin C} = \frac{[\cos(-A) + i \sin(-A)](\cos B + i \sin B)}{\cos(-C) + i \sin(-C)}$$

$$= \cos[-A + B - (-C)] + i \sin[-A + B - (-C)] = \cos(B + C - A) + i \sin(B + C - A) \in \mathbb{R}$$

$$\therefore \sin(B + C - A) = 0 \Rightarrow B + C - A = 0^\circ, \text{而 } B + C + A = 180^\circ \Rightarrow A = 90^\circ \Rightarrow \text{直角三角形}$$

$$19. \beta^2 - 2\alpha\beta + 4\alpha^2 = 0 \Rightarrow (\alpha - \beta)^2 + 3\alpha^2 = 0 \quad \therefore \alpha - \beta = \pm\sqrt{3} \cdot i \cdot \alpha = \pm\sqrt{3} \cdot (\cos 90^\circ + i \sin 90^\circ) \cdot \alpha$$

$$\text{可知, } \alpha - \beta \text{ 爲 } \alpha \text{ 順/逆時針轉 } 90^\circ, \text{長度爲 } \alpha \text{ 之 } \sqrt{3} \text{ 倍 } \therefore |\alpha| = 1, \text{所求 } 1 \cdot \sqrt{3} \cdot \frac{1}{2} = \frac{\sqrt{3}}{2}$$

P.23

$$20. (\sqrt{3} - i)^n = 2^n \cdot \left(\frac{\sqrt{3}}{2} - \frac{1}{2}i \right)^n = 2^n \cdot (\cos 330^\circ + i \sin 330^\circ)^n = 2^n \cdot [\cos(330^\circ \cdot n) + i \sin(330^\circ \cdot n)]$$

$$(1 + \sqrt{3}i)^n = 2^n \cdot \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right)^n = 2^n \cdot (\cos 60^\circ + i \sin 60^\circ)^n = 2^n \cdot [\cos(60^\circ \cdot n) + i \sin(60^\circ \cdot n)]$$

$$\therefore (\sqrt{3} - i)^n = (1 + \sqrt{3}i)^n = 2^n \cdot \{ \cos(330^\circ \cdot n) + \cos(60^\circ \cdot n) + i [\sin(330^\circ \cdot n) + \sin(60^\circ \cdot n)] \}$$

$$\therefore \begin{cases} \cos 330^\circ \cdot n + \cos 60^\circ \cdot n = 0 \\ \sin 330^\circ \cdot n + \sin 60^\circ \cdot n = 0 \end{cases}$$

$$\text{同時滿足 } \begin{cases} \cos \text{ 差負號表兩角差 } 180^\circ + 360^\circ k \text{ 或兩角和 } 180^\circ + 360^\circ k \\ \sin \text{ 差負號表兩角差 } 180^\circ + 360^\circ k \text{ 或兩角和 } 360^\circ k \end{cases}$$

$$\therefore 330^\circ n - 60^\circ n = 270^\circ n = 180^\circ + 360^\circ k \Rightarrow n = 2, 6, 10, 14, \dots$$

$$n \text{ 值和 } = (2 + 98) \times 25 \div 2 = 1250 \quad \text{答: (1)最小2 (2)和1250}$$

$$21. 2x - 1 + \sqrt{3}i = 0 \Rightarrow (2x - 1)^2 = (-\sqrt{3}i)^2 \Rightarrow 4x^2 - 4x + 1 = -3 \Rightarrow x^2 - x + 1 = 0 \Rightarrow x^3 + 1 = 0, (x \neq 1)$$

$$\text{所求} = 1 + x + x^2 + \cdots + x^{12} = (1 + x^3 + x^6 + x^9 + x^{12}) + (x + x^4 + x^7 + x^{10}) + (x^2 + x^5 + x^8 + x^{11})$$

$$(1 - 1 + 1 - 1 + 1) + (x - x + x - x) + (x^2 - x^2 + x^2 - x^2) = 1$$

$$22. (1 + \cos 20^\circ + i \sin 20^\circ)^{987} = \left[1 + (2 \cos^2 10^\circ - 1) + i 2 \sin 10^\circ \cos 10^\circ \right]^{987} \\ = (2 \cos 10^\circ)^{987} (\cos 10^\circ + i \sin 10^\circ)^{987} = \underbrace{(2 \cos 10^\circ)^{987}}_r (\cos 9870^\circ + i \sin 9870^\circ), \theta = 150^\circ$$

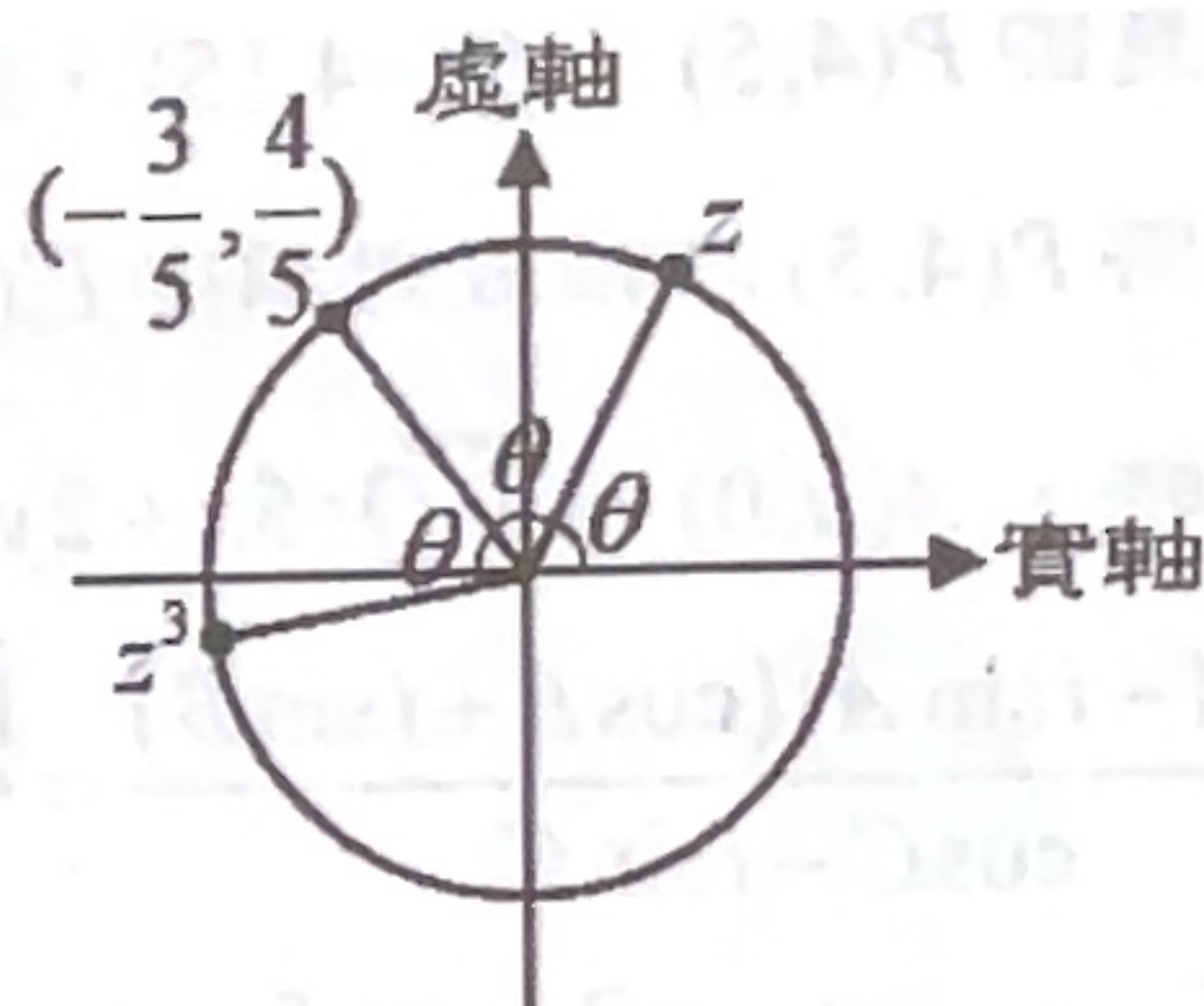
$$23. \text{由 } (Z^2 - 1)(Z^6 + Z^4 + Z^2 + 1) = Z^8 - 1 = 0, Z^8 = 1 \Rightarrow Z^8 = r^8 (\cos 8\theta + i \sin 8\theta)$$

$$\begin{cases} r^8 = 1 \\ 8\theta = 2k\pi, k \text{ 爲整數} \end{cases} \Rightarrow \begin{cases} r = 1 \\ \theta = \frac{k}{4}\pi, k \text{ 爲整數} \end{cases}, Z = 1(\cos \frac{k\pi}{4} + i \sin \frac{k\pi}{4}), k = 0 \sim 7$$

$$Z^2 - 1 = 0 \text{ 之解不是原方程式解, 所求} = 4 \cdot \frac{1}{2} \cdot 1 \cdot 1 \cdot \sin 45^\circ + 4 \cdot \frac{1}{2} \cdot \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = 2 \cdot \frac{\sqrt{2}}{2} + 1 = \frac{\sqrt{2} + 1}{2}$$

$$24. \left| \frac{-3+4i}{5} - z^3 \right| = \left| \frac{-3+4i}{5} - z \right| \text{ 即爲複數平面上}$$

$(-\frac{3}{5}, \frac{4}{5})$ 到 z^3 的距離 = $(-\frac{3}{5}, \frac{4}{5})$ 到 z 的距離, 可作圖:



[法 I] 設 $z = \cos \theta + i \sin \theta$, 則 $z^2 = \frac{-3+4i}{5}$

$$z = \pm \sqrt{\frac{-3+4\sqrt{-1}}{5}} = \pm \sqrt{\frac{-3+2\sqrt{-4}}{5}}$$

$$= \pm \frac{\sqrt{1+\sqrt{-4}}}{\sqrt{5}} = \pm (\frac{\sqrt{5}}{5} + \frac{2\sqrt{5}}{5}i) \text{ (取正), 故 } a = \frac{\sqrt{5}}{5}, b = \frac{2\sqrt{5}}{5}$$

[法 II] 設 $z = \cos \theta + i \sin \theta$, 則 $-\frac{3}{5} + \frac{4}{5}i = \cos 2\theta + i \sin 2\theta \Rightarrow \begin{cases} \cos 2\theta = -\frac{3}{5} \\ \sin 2\theta = \frac{4}{5} \end{cases}$

$$\text{所求 } a = \cos \theta = \sqrt{\frac{1 + \cos 2\theta}{2}} = \sqrt{\frac{1 + (-\frac{3}{5})}{2}} = \frac{\sqrt{5}}{5}$$

$$b = \sin \theta = \sqrt{\frac{1 - \cos 2\theta}{2}} = \sqrt{\frac{1 - (-\frac{3}{5})}{2}} = \frac{2\sqrt{5}}{5}$$

$$25. (1) z^3 = (2i)^3 = -8i, 4i\bar{z} = 4i \cdot (-2i) = -8, \text{故當 } z = 2i \text{ 時, } z^3 \neq 4i\bar{z}$$

$$(2) |\alpha|^3 = |4i\bar{\alpha}| = 4|\alpha| \Rightarrow |\alpha| = 2$$

利用(2), 設 $z = 2(\cos \theta + i \sin \theta)$ 代入 $z^3 = 4i\bar{z}$,

$$\text{有 } 8(\cos 3\theta + i \sin 3\theta) = 4(\cos(\frac{\pi}{2}) + i \sin(\frac{\pi}{2})) \cdot 2(\cos(-\theta) + i \sin(-\theta))$$

$$\text{可得 } \cos 4\theta + i \sin 4\theta = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \Rightarrow \theta = \frac{\frac{\pi}{2} + 2k\pi}{4}, k = 0, 1, 2, 3$$

(3) 若 α 滿足條件, 表示 αi (旋轉 90°) 亦滿足條件 (4) 最小角度爲 $\frac{\pi}{8}$

(5) 有四組相異的複數 z 滿足條件